

*The Introduction chapter provides a **clear overview of the project**, explains **why the problem is important**, and defines **what the project aims to achieve**, without going into detailed technical analysis.*

Chapter One Sections and How to Write Them

1. Background of the Study

What to include:

- General overview of the engineering field related to the project
- Technological development and current trends
- Common real-world applications

How to write:

- Start from general concepts and gradually narrow to the specific topic
- Use simple, clear academic language
- Avoid equations and technical design details
- Rely on textbooks and authoritative review sources

Example:

In recent years, electrical engineering systems have experienced significant technological advancements, particularly in the areas of automation and intelligent control...

2. Problem Statement

What to include:

- Clear description of the engineering problem
- Limitations or shortcomings of existing solutions
- Justification for addressing this problem

How to write:

- Be concise and specific
- Avoid vague or overly broad descriptions
- Clearly link the problem to practical engineering challenges

3. Significance of the Project

What to include:

- Academic or educational importance
- Practical or industrial relevance
- Expected benefits or improvements

How to write:

- Use short paragraphs or bullet points
- Relate significance directly to the identified problem

4. Project Objectives

What to include:

- One general objective
- Several specific objectives

How to write:

- Use clear engineering action verbs such as:
Design, Implement, Analyze, Simulate, Evaluate
- Ensure objectives are measurable and achievable

Example:

- To design an efficient ...
- To simulate the proposed system using ...
- To evaluate system performance under different conditions

5. Scope and Limitations of the Study

What to include:

- What the project covers
- What is excluded from the project
- Technical, time, or resource constraints

How to write:

- Clearly define boundaries to manage expectations
- Keep the section brief and precise

6. Methodology Overview

What to include:

- General steps followed to complete the project

How to write:

- High-level description only
- No equations or detailed algorithms

Example:

The methodology begins with a theoretical study, followed by system design, simulation, hardware implementation, and performance evaluation.

7. Organization of the Thesis

What to include:

- Brief description of each chapter

How to write:

- One concise paragraph
- Follow the actual chapter sequence

Chapter Two: Theoretical Background and Literature Review

*This chapter establishes the **scientific foundation** of the project and demonstrates the student's understanding of existing theories, technologies, and related research.*

Chapter Two Sections and How to Write Them

Part I: Theoretical Background

1. Fundamental Concepts

What to include:

- Definitions of key terms
- Core principles and theories
- Basic mathematical relationships (if necessary)

How to write:

- Use an educational, explanatory tone
- Explain equations in words
- Include diagrams where helpful

2. Related Systems and Models

What to include:

- Description of existing systems or architectures
- Functional blocks and their roles

How to write:

- Progress from general system descriptions to specific ones
- Relate each component to its function in the project

3. Techniques, Methods, or Algorithms

What to include:

- Explanation of techniques used in the project
- Operating principles and advantages

How to write:

- Avoid copying directly from sources
- Briefly mention alternatives if relevant

Part II: Literature Review

4. Review of Previous Studies

What to include:

- Relevant journal papers, conference articles, or similar projects “thesis”
- Research objectives and methodologies
- Key findings

How to write:

- Organize studies (15 – 25) chronologically or thematically
- Summarize, do not reproduce, the original work

Example:

Author (Year), title, proposed a system based on ... The results showed that ...

5. Comparative Analysis of Previous Work

What to include:

- Comparison between existing studies
- Strengths and weaknesses of each approach

How to write:

- Use tables where possible
- Focus on technical relevance to your project

6. Research Gap and Project Contribution

What to include:

- Limitations or gaps in existing research
- How the current project (yours) will address these gaps.

How to write:

- Strong concluding paragraph
- Clearly connect the gap to the project objectives

Important Notes

- Avoid plagiarism; always paraphrase and cite sources
- Only include theory that supports your project
- Use recent references where possible (last 5–10 years)
- Follow a consistent referencing style (IEEE is commonly used)